



# Prayas JEE 2.0 (2024)

## Quadratic Equations

**DPP-07**

- Range of the expression  $\frac{16x^2 - 12x + 9}{16x^2 + 12x + 9} : (x \in R)$  is-
  - $\left[\frac{1}{3}, 3\right]$
  - $\left(-\infty, \frac{1}{3}\right]$
  - $[3, \infty)$
  - $R$
- If exactly one root of the equation  $2^k x^2 - 4^k x + 2^k - 1 = 0$  lies in  $[0, 1)$ , then complete range of  $k$  is-
  - $(-\infty, 0]$
  - $(-\infty, 0)$
  - $(0, \infty)$
  - $[0, \infty)$
- If  $\alpha, \beta$  are the roots of  $x^2 - 3x + a = 0$ ,  $a \in R$  and  $\alpha < 1 < \beta$  then
  - $a \in (-\infty, 2)$
  - $a \in \left(-\infty, \frac{9}{4}\right]$
  - $a \in \left[2, \frac{9}{4}\right]$
  - none of these
- The values of  $a$ , for which the quadratic equation  $3x^2 + 2(a^2 + 1)x + (a^2 - 3a + 2) = 0$  possesses roots of opposite sign, are
  - $1 < a < 2$
  - $a \in (2, \infty)$
  - $1 < a < 3$
  - $-1 < a < 0$
- If both the roots of the quadratic equation  $x^2 - 2kx + k^2 + k - 5 = 0$  are less than 5, then  $k$  lies in the interval
  - $(5, 6]$
  - $(6, \infty)$
  - $(-\infty, 4)$
  - $[4, 5]$
- All the values of  $m$  for which both roots of the equation  $x^2 - 2mx + m^2 - 1 = 0$  are greater than  $-2$  but less than 4, lie in the interval
  - $-2 < m < 0$
  - $m > 3$
  - $-1 < m < 3$
  - $1 < m < 4$
- The set of all real values of  $\lambda$  for which the quadratic equation,  $(\lambda^2 + 1)x^2 - 4\lambda x + 2 = 0$  always have exactly one root in the interval  $(0, 1)$  is
  - $(-3, -1)$
  - $(2, 4]$
  - $(0, 2)$
  - $(1, 3]$
- If  $0 < a < b < c < d \leq 1$ , equation  $ax^2 + (1 - ab - ac)x + abc - d = 0$  has
  - real roots
  - roots of opposite signs
  - one root  $\in (a, b)$
  - one root  $\in (b, c)$
- If the inequality  $x^2 + ax + a^2 + 6a < 0$  is satisfied for all  $x \in (1, 2)$ , then the sum of all the integral values of  $a$  must be equal to
  - $-10$
  - $-15$
  - $-21$
  - $-28$
- If the equation  $ax^2 + bx + c = 0$  ( $a > 0$ ) has two real roots  $\alpha$  and  $\beta$  such that  $\alpha < -2$  and  $\beta > 2$ , which of the following statements is/are true? **[Challenger]**
  - $4a - 2|b| + c < 0$
  - $9a - 3|b| + c < 0$
  - $a - |b| + c < 0$
  - $c < 0, b^2 - 4ac > 0$



**Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.**

## Answer Key

1. (1)
2. (3)
3. (1)
4. (1)
5. (3)

6. (3)
7. (4)
8. (1, 2)
9. (3)
10. (1, 3, 4)



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